CAPSTONE PROJECT · CLASS 21PFIVEV2 · FAST – DUT

DESIGN AND IMPLEMENTATION OF AN IOT-BASED INDOOR AIR QUALITY MONITORING SYSTEM FOR EDUCATIONAL ENVIRONMENTS

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AGENDA

What We'll Cover

Why is it necessary to monitor air quality in classrooms?

Figure 1: Air Pollution in a Meeting Room
https://tapchimoitruong.vn/

IMPACT ON HEALTH
Poor air quality can cause headaches,
fatigue ,respiratory problems and other health issues

IMPACT ON LEARNING
High CO_2 levels and pollutants reduce concentration,
memory and overall learning performance.

NEED FOR MONITORING
Many classrooms lack continuous monitoring.
making it difficult to detect air quality problems.

REAL-TIME MONITORING
Real- time monitoring helps detect problems early
and allows timely actions to improve air quality

Research Objectives

Design a smart air quality monitoring system for classrooms

- Real-time IAQ monitoring
- Early warning when CO_2 and PM2.5 exceed safety thresholds
- Provide data for trend analysis
- All collected data are uploaded to a cloud dashboard
- Support classroom environment management
- Increase awareness among students and teachers

Figure 2: Threshold Exceedance Warning

Overview of indoor air quality in classrooms

1.PARTICULATE MATTER(PM2.5,PM10)
Fine particles that can respiratory system, affect healthy

2.VOC/TVOC
Gases released from building materials, furniture, cleaning products and daily activities

3.CO2
Which reflects the ventilation level of the classroom

4. TEMPERATURE
Affects thermal comfort, ability to concentrate and overall learning performance

5. HUMIDITY
Too high or too low humidity can promote mold, allergens, irritation

6.OTHER GASES (NO2,SO2)
Can cause respiratory irritation and affect long – term health

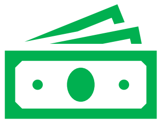
Commercial product

Figure 3: Temtop LKC-1000S Portable Air Quality Monitor

Figure 4: IQAir AirVisual Pro Air Quality Sensor Device

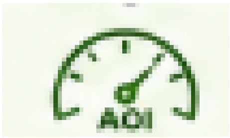
Figure 5: Xiaomi PM2.5 Air Quality Detector

OUR PROPOSED SOLUTION


Low cost

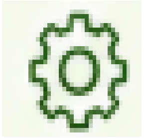

Multi-parameter


Cloud database



AQI calculation


Real-time dashboard


Customizable software

General Introduction to Air Quality Standards

MAIN REGULATIONS

- ✓ QCVN and TCVN environmental standards
- ✓ Issued by the Ministry of Natural Resources and Environment
- ✓ Circular No. 10/2021/TT-BTNMT

DEVICE REQUIREMENTS

- ✓ Automatic and continuous measurement
- ✓ Real-time data transmission
- ✓ Periodic verification and calibration
- ✓ Ensure accuracy and appropriate measurement range

Pollution Level Classification According to IAQ

IAQ Range	Pollution Level	Display Color	Health Impact
0 – 50	Good	Green	No health effects
51 – 100	Moderate	Yellow	May cause minor effects for sensitive individuals
101 – 150	Unhealthy for Sensitive Groups	Orange	Affects sensitive groups(elderly, children)
151 – 200	Unhealthy	Red	Affects everyone’s health
201 – 300	Very Unhealthy	Purple	Serious health effects
> 300	Hazardous	Maroon	Emergency health warning

Table 2: Air Quality Assessment Levels

Technical specification requirements

No	Measurement Parameter	Unit	Accuracy(% of range)	Measurement Range	Resolution	Response Time
1	Temperature	°C	±5%	0 ÷ 80	0,1	≤ 120 seconds
2	PM2.5 dust	µg/m³	±5%	0 ÷ 250	0,1	≤ 60 seconds
3	PM10 dust	µg/m³	±3%	0 ÷ 500	0.1	≤ 60 seconds
4	CO2	ppm	±5%	0 ÷ 5.000	1	≤ 120 seconds
5	TVOC	µg/m³	±10%	0 ÷ 2.000	1	≤ 60 seconds

Table 3: Required Technical Specifications for Measured Parameters

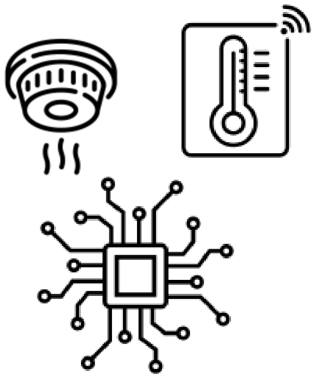
Ensures accuracy and reliability of measurement

Meets national standards for IAQ monitoring

Real-time response for timely alerts and actions

Overall System Architecture

DEVICE LAYER

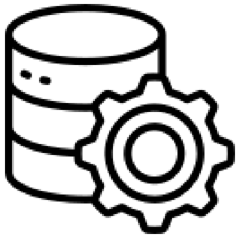


Environmental Sensor Data Acquisition and process signals

NETWORK LAYER

Security Protocol HTTPS/TLS (Wifi ESP32)

SERVICE LAYER

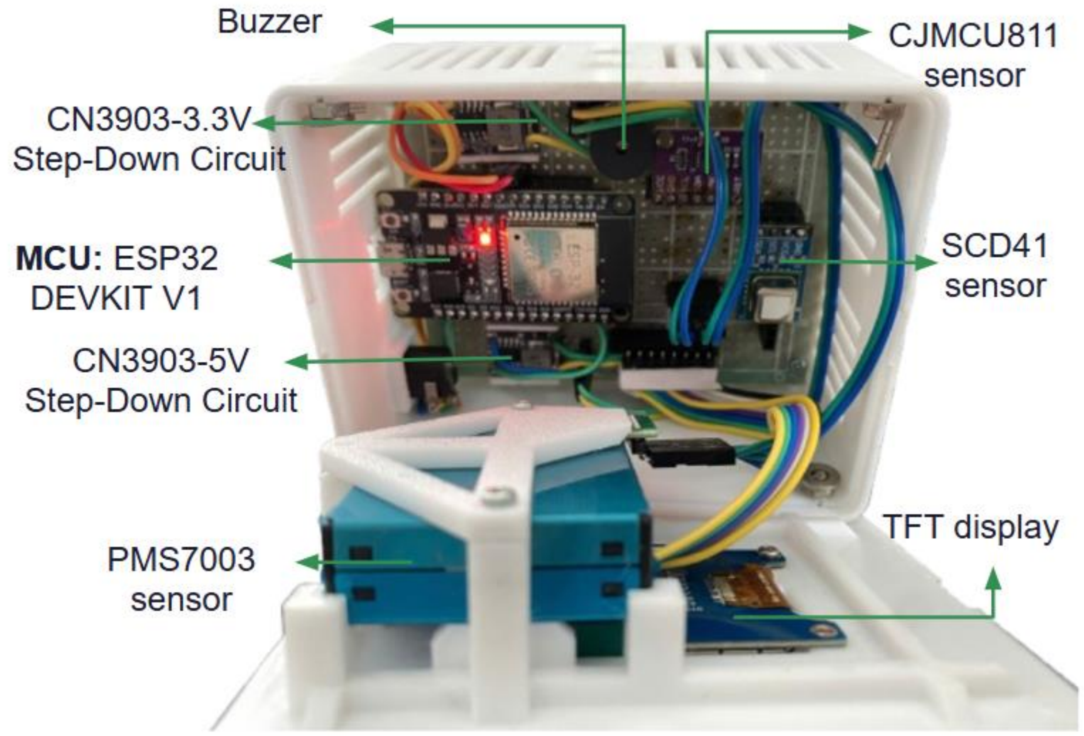


Real-Time Data Storage and Processing (Supabase PostgreSQL, RPC)

APPLICATION LAYER

Real-Time Data Visualization and Monitoring (React)

Hardware Block Diagram & Circuit Schematic



ESP32 DevKit V1 (Concentrator)

Xtensa LX6 dual-core, 240 MHz, 520 KB SRAM · 4 MB Flash
Wi-Fi b/g/n · BT 4.2, FreeRTOS multi-task firmware

SCD41 (CO2, Temperature, Humidity)

Photoacoustic CO_2 sensing · Integrated Temp & Humidity · I^2C Interface (0x62)

PMS7003 (Dust PM2.5, PM10)

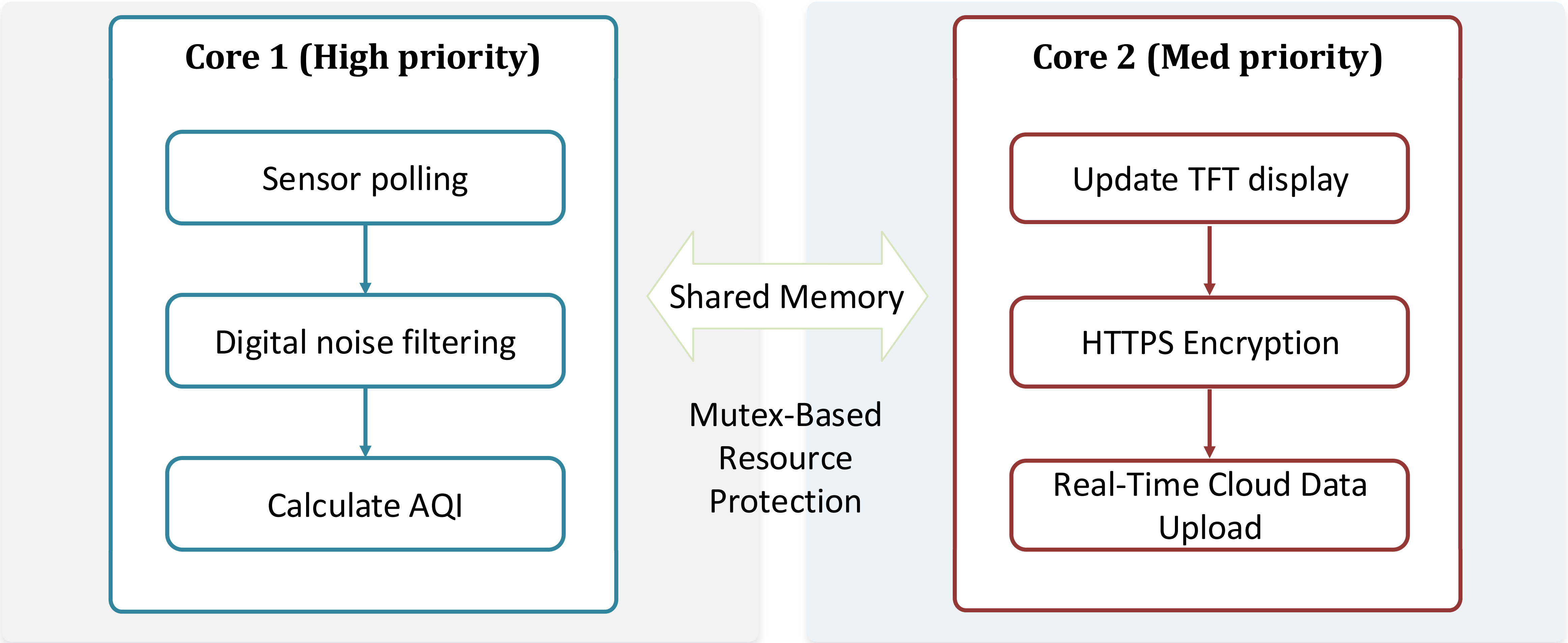
Laser Scattering Technology · Particle Measurement · UART Interface

CCS811 (TVOC)

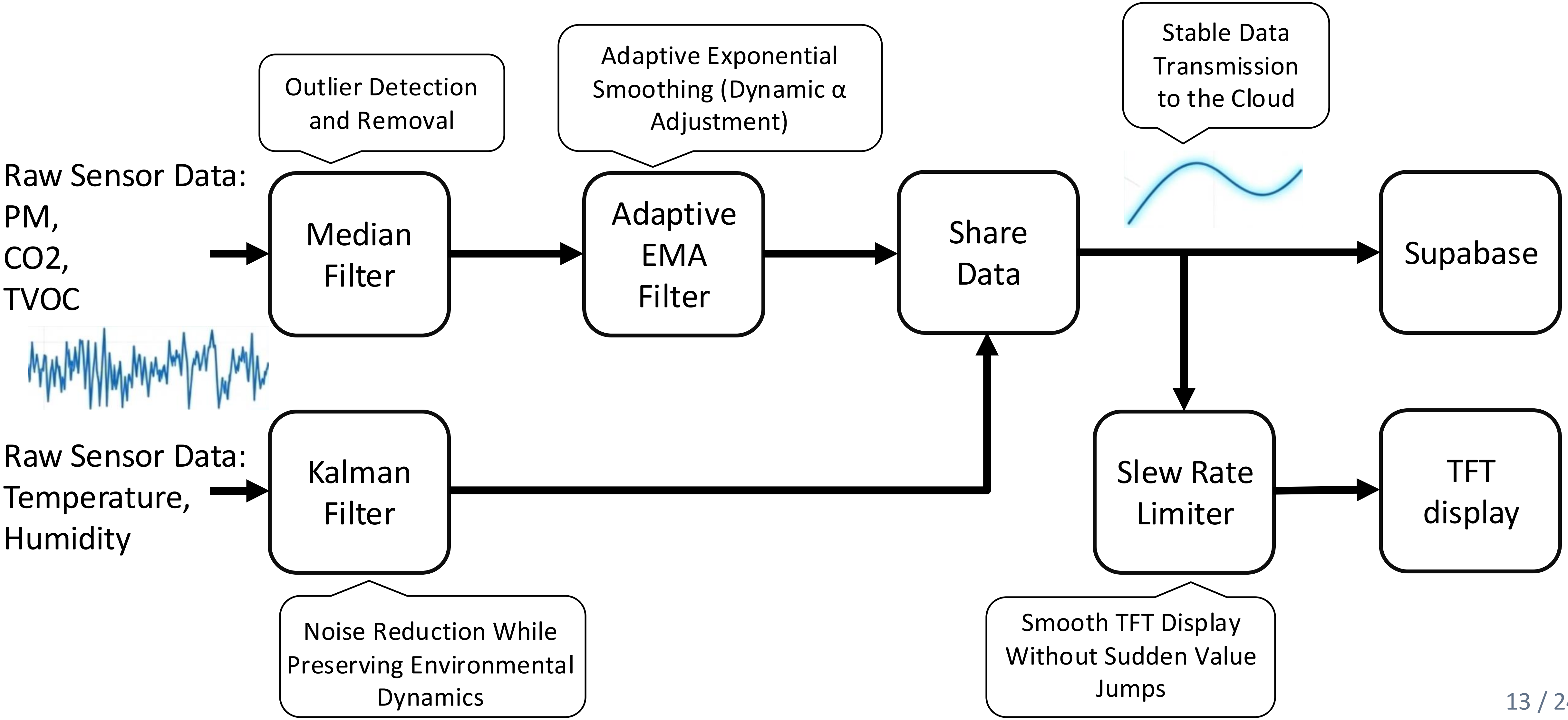
MOX Gas Sensing · TVOC Estimation · I^2C Interface (0x5A)

Multitasking with FreeRTOS on a Dual-Core Architecture

Key Feature: Wi-Fi Network Latency Never Interrupts the Sensor Reading Cycle



Signal Processing Pipeline



Cloud Storage Architecture (Supabase/PostgreSQL)

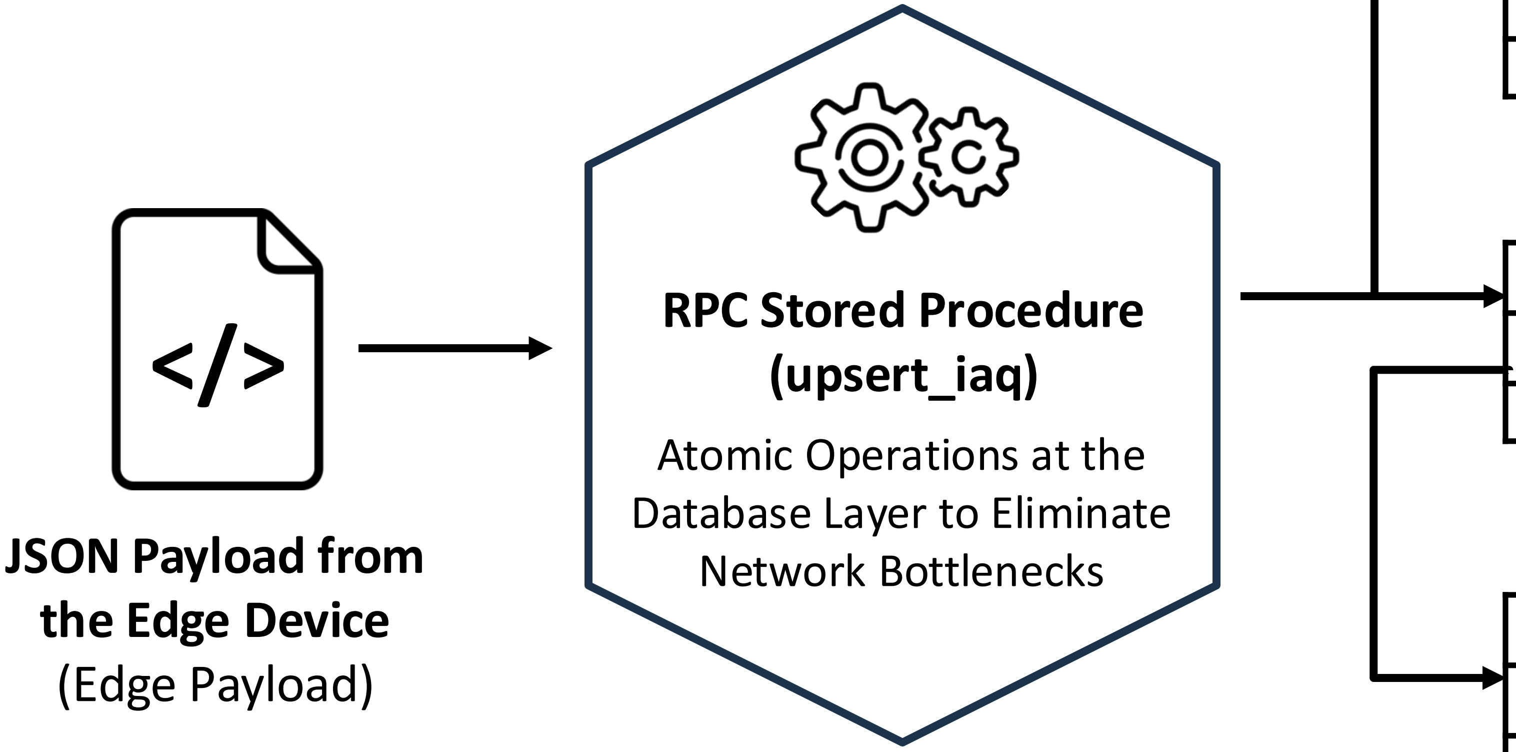


Table aqi_room

Schema				
id	room_id	aqi	temp	…
…	…	…	…	

Latest-State Table for Fast User Interface Response

Table aqi_room_history

Schema				
id	room_id	aqi	temp	…
…	…	…	…	

Time-Series Log for Continuous Data Archiving

Table aqi_hourly

Schema				
id	room_id	aqi	temp	…
…	…	…	…	

Hourly Aggregated Data for Efficient Long-Term Visualization

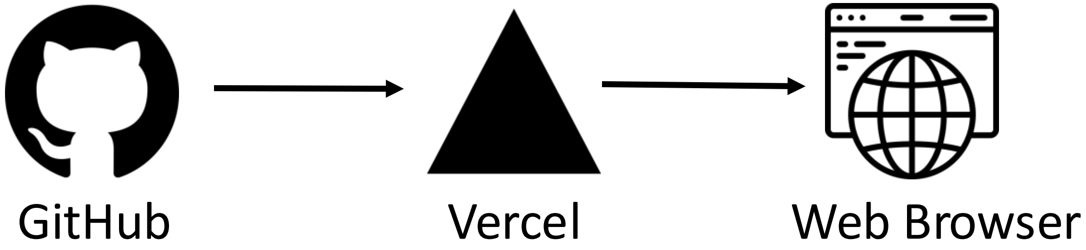
Web Monitoring Interface

Technologies: React 19 + Vite + Chart.js

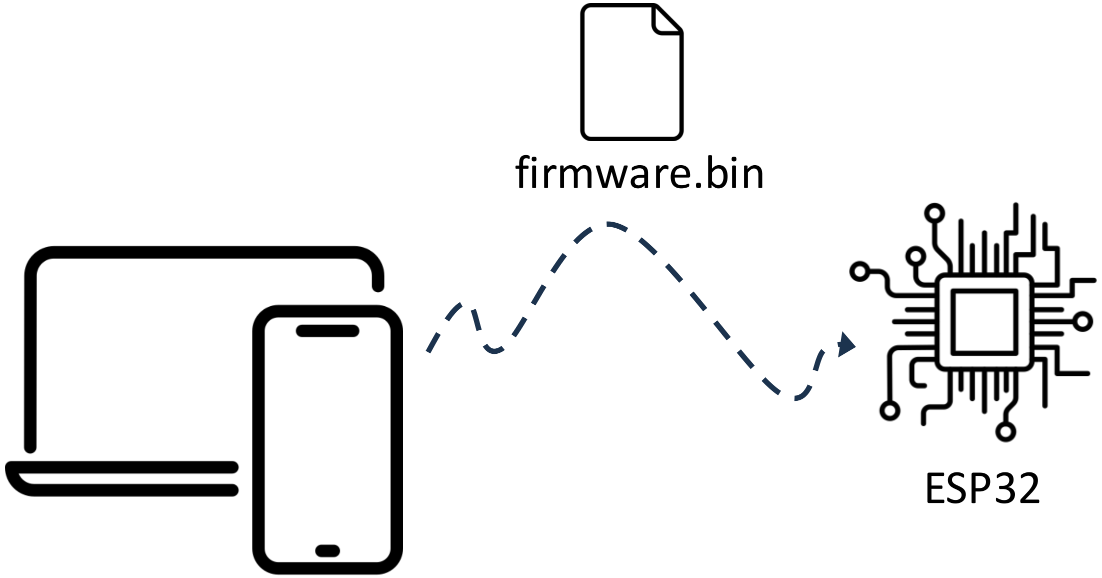
Real-time Sync: Real-Time Data Updates Using Supabase WebSocket

Deployment: Automated CI/CD Deployment via GitHub and Vercel

Key Features: Multi-Room Monitoring, Interactive Historical Charts, Real-Time Visual Alerts



Wi-Fi Captive Portal & update OTA



Update Over-The-Air (OTA)

Fail-Safe Dual-Partition OTA (App0/App1): Prevents Firmware Corruption During Interrupted Updates

Wifi manager

Built-In Configuration Portal: Easily Set Up Wi-Fi, Room ID, and Room Name Without Modifying the Source Code

Experimental Environment and Configuration

Experimental Configuration Parameters

Parameter	Value
1 Sampling period	10 seconds
2 Measurement duration	12 hours (8:30 am – 8:30 pm)
3 Total number of samples	~ 4.320
4 Protocol	HTTPS
5 AQI	US EPA

Practical Deployment Information

📍 *Location: Classroom room 5 (PFIEV)*

👥 *Human density: Fluctuates between 20 – 40 students*

Installation location: Room center, 1.2m high, away from window and air conditioners

Evaluation of Operational Capability

The system operated continuously for 12 hours, the results achieved

Metric	Value
Uptime	12 hours
Packet Success Rate	98.6%
Average Latency	0.98s
Min Latency	0.91s
Max Latency	1.05s
System Crash	0
Sensor Communication Failure	0
Wi-Fi Reconnection Success Rate	100%

The system well meets the requirements for reliability and stability inreal-world environments

Figure 6: Real-time air quality monitoring Dashboard interface

Evaluation of System Sensor Errors

Measurement Parameter	MAE Index	MAPE Index	Quality Evaluation	Conclusion
Temperature	0.95 °C	2.85%	Extremely small error, completely tracking the national standard measurement device	Very high
PM2.5	1.19 µg/m³	8.31%	Lies below the 10% error threshold, extremely accurately reflecting the variation of fine dust concentration	Good
PM10	24.59 µg/m³	53.49%	Fairly high error, measured values are always lower than reality by about 2 times. Due to large dust particles easily settling and the influence of airflow structure	Average (Recalibration needed)
Humidity	7.43%RH	12.51%	Tends to measure 8-10% lower than the standard in high humidity ranges, but still ensures high stability	Stable

Table 4: Accuracy evaluation compared to the reference monitoring station

Actual CO_2 & TVOC Fluctuations

Figure 7: CO_2 concentration variation over time

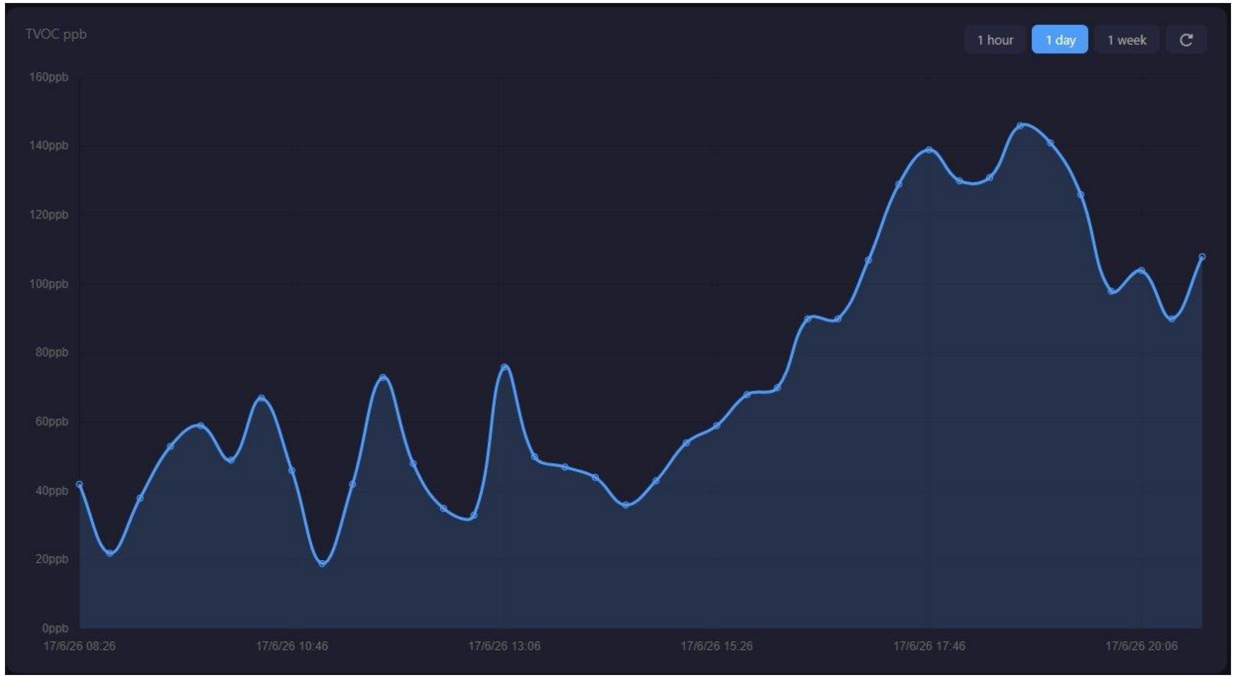


Figure 8: TVOC concentration variation over time

CO_2 Carbon Dioxide Gas

- Fluctuated most strongly according to human behavior and student density, peaking at 900 - 1000 ppm during peak class hours
- Decreased rapidly to a stable range of 450 - 550 ppm when opening doors for ventilation or after class dismissed, reflecting the effectiveness of natural ventilation

TVOC Organic Compounds

- Fluctuated within the range of 20 - 150 ppb. Accumulated with a delay and surged suddenly to peak at the end of the measurement session.
- Continuously generated from the plywood adhesive of tables and chairs, classroom cleaning activities, and heat dissipated from computer devices.

Actual PM2.5 & AQI Fluctuations

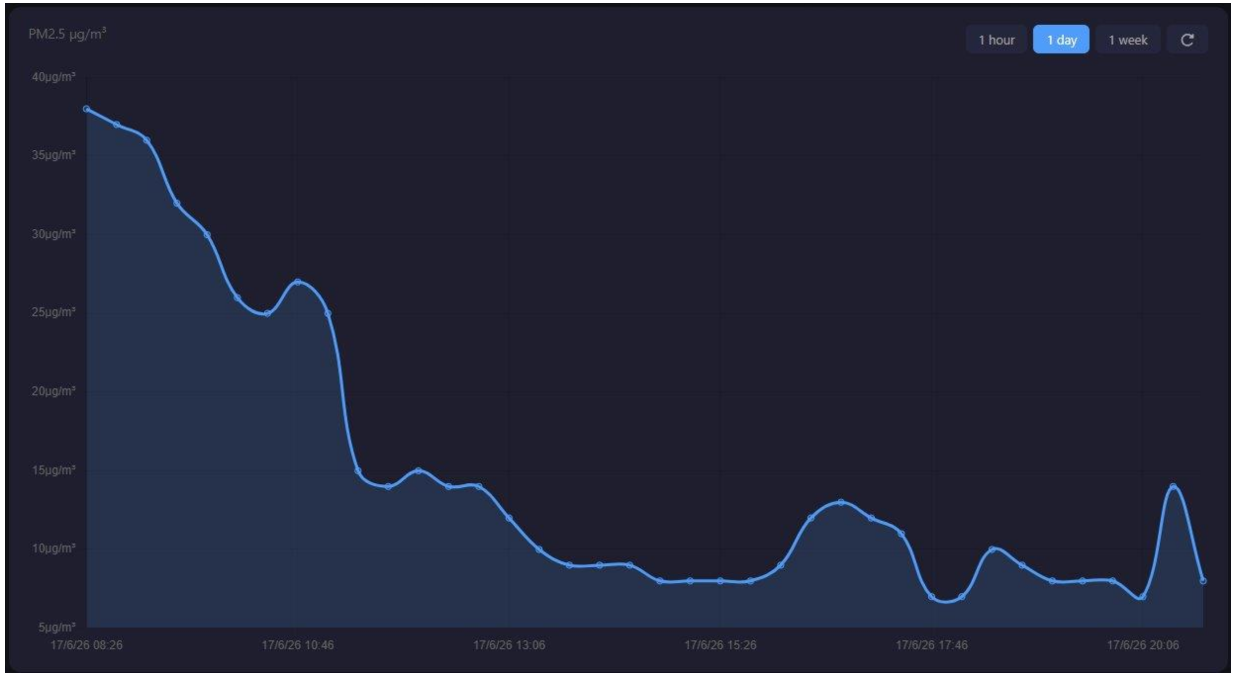


Figure 9: PM2.5 concentration variation over time

Figure 10: AQI index variation over time

PM2.5 Fine Dust Concentration

- Gradually decreased from the initial level of 38 - 40 µg/m³ and maintained extremely good stability within the safe range of 5 - 15 µg/m³
- Due to the classroom doors being closed, effectively preventing the infiltration of dust generated from outdoor traffic flow

Composite AQI Index

- Clear downward trend from 105 - 110 (start of shift) to an ideal level of 30 - 40 (end of shift)
- Met the "Good" standard according to the US EPA most of the time; despite having localized sharp peaks (50 - 55), it recovered quickly

General Conclusion

Objectives

Successfully completed the design and fabrication of the IoT-based multi-parameter monitoring system, perfectly matching all initial project specifications.

Technical

Effectively integrated ESP32 microcontrollers, Supabase cloud databases, and specialized precision sensors to achieve stable, high-accuracy data ingestion.

Academic

Rigorously validated system error rates (MAE, MAPE) against certified standards, establishing acceptable precision thresholds for civilian air monitoring.

Practicality

Developed a highly viable, cost-optimized solution readily deployable across smart classrooms and next-generation green school environments.

Future Development Directions

🛠 01. Hardware & Infrastructure Optimization

- Integrate additional specialized toxic gassensors (HCHO, CO, NO_2)
- Deploy a remote update mechanism via OTA and synchronize centralized managementacross multiple classrooms

💻 02. Software & Algorithms

- Apply AI/ML to predict air quality
- Develop a Mobile App to receive instantalerts
- Interconnected control of the automaticventilation system

THANK YOU FOR LISTENING

Q & A

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